

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \rightarrow PA$ (8.1).

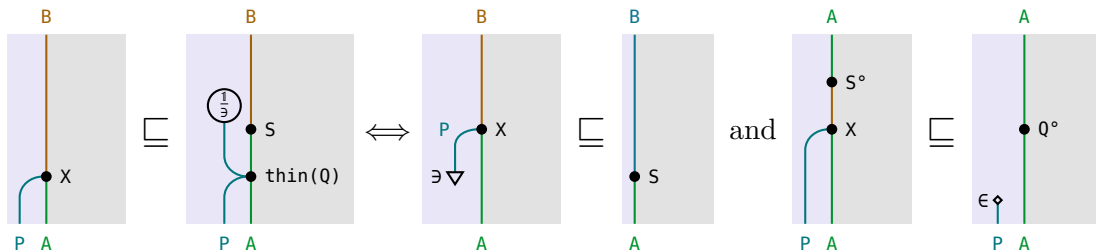
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q \triangleq \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

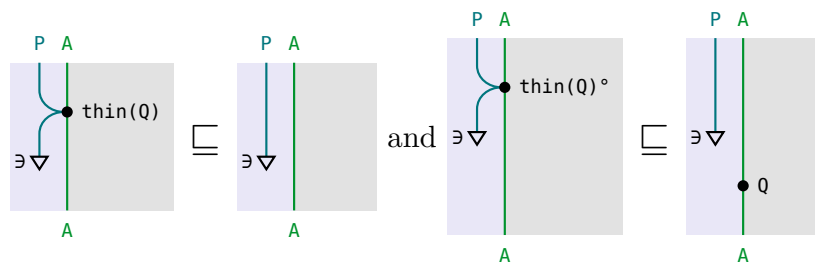
the law	what it says
1 $X \in \frac{S}{\exists} \text{ thin}(Q) \iff X \exists \in S \text{ and } S^\circ X \in Q^\circ \epsilon$	everything kept is an S-value, and every S-value has a Q-lower bound among the kept ones
2 $\text{thin}(Q) \exists \in \exists$	everything a thinning keeps was in the set
3 $\text{thin}(Q)^\circ \exists \in \exists \ Q$	every element of the set has a Q-lower bound among the kept ones
4 $Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
5 $1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \subseteq Q \implies \text{thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
6 $Q \subseteq R \wedge 1 \subseteq Q \wedge R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R)$ $Q \subseteq R, 1 \subseteq Q, R \subseteq R \implies (8.1.2a) \ 1 \subseteq Q$ $\implies \text{est}(R) \subseteq \text{thin}(Q) \text{ est}(R)$; weaker than the book's "both preorders": Q transitive is never used and $1 \subseteq R$ follows	thin-introduction : thinning first cannot lose an R-minimum
7 $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists} \quad (8.2) \implies (8.2) \text{ est}(Q) \frac{1}{\exists} \subseteq \text{thin}(Q)$	thin-elimination : keeping one element is a thinning, but its domain is the sets est (Q) is defined on
8 $\frac{S}{\exists} \text{ thin}(Q) \supseteq \frac{S}{\exists} \text{ est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
9 $P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q) \quad (8.4)$	thinning each member set is a thinning of the union

§8.1.1. $X \in (S/\exists) \text{ thin}(Q) \iff X \exists \in S \text{ and } S^\circ X \in Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)

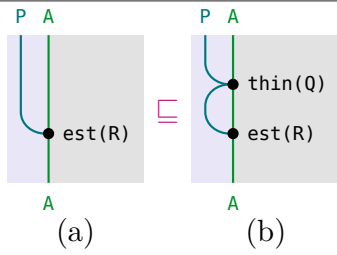


(8.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \exists = \wedge(\exists) = 1$

§8.1.2. $\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ given $Q \subseteq R, 1 \subseteq Q, R \subseteq R$

$$1 \sqsubseteq Q \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R) \quad (8.1.2a)$$

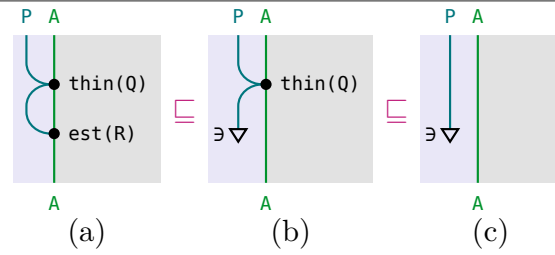
the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$



(a)	$\text{est}(R)$	$1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$ $\Rightarrow 1 \sqsubseteq \text{thin}(Q)$, $Q \sqsubseteq Q$ $\Rightarrow \text{thin}(Q)$ $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$
(b)	$\sqsubseteq \text{thin}(Q) \text{ est}(R)$	

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

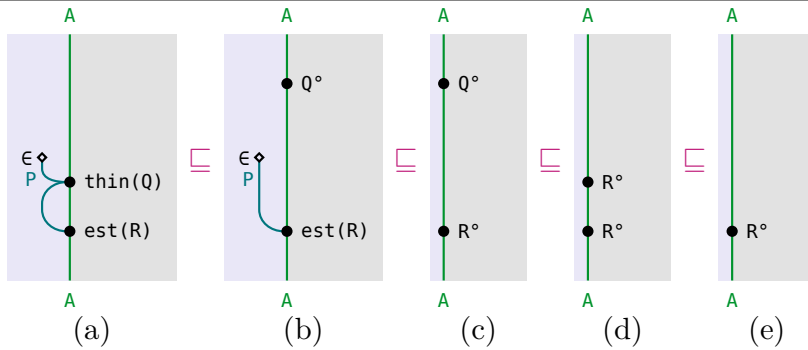
the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — (7.1.1a)



(a)	$\text{thin}(Q) \text{ est}(R)$	$R \cap S \sqsubseteq R$
(b)	$\sqsubseteq \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(c)	$\sqsubseteq \exists$	

$$Q \sqsubseteq R \wedge R \sqsubseteq R \Rightarrow \epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive



(a)	$\epsilon \text{ thin}(Q) \text{ est}(R)$	$\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon$
(b)	$\sqsubseteq Q^\circ \epsilon \text{ est}(R)$	$\epsilon \text{ est}(R) \sqsubseteq R^\circ$
(c)	$\sqsubseteq Q^\circ R^\circ$	$Q \sqsubseteq R$
(d)	$\sqsubseteq R^\circ R^\circ$	$R \sqsubseteq R$ $\Rightarrow R^\circ R^\circ \sqsubseteq R^\circ$
(e)	$\sqsubseteq R^\circ$	

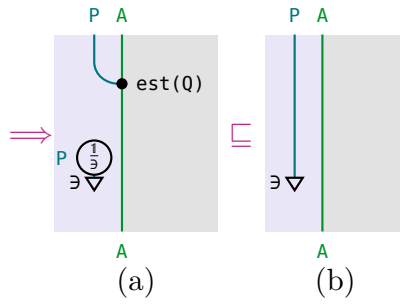
$$\text{est}(Q) \stackrel{1}{\exists} \sqsubseteq \text{thin}(Q)$$

(8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$$\text{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

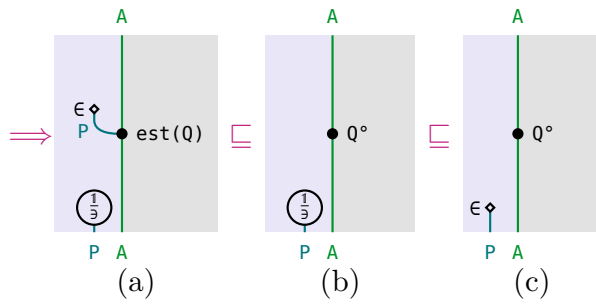
every member of the singleton is a member of the set



(a)	$\Rightarrow \text{est}(Q) \stackrel{1}{\exists} \ni$	$\stackrel{1}{\exists} \ni = \mathbb{1} \text{ — } \stackrel{R}{\exists} \ni = R \text{ est}(Q) \sqsubseteq \ni \text{ — } \triangleq \text{est}$
(b)	$\sqsubseteq \ni$	$\triangleq \text{thin}, \epsilon \backslash (Q^\circ \epsilon) \text{ half}$

$$\epsilon \text{ est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \epsilon$$

the singleton's member stands in Q to every member of the set



(a)	$\Rightarrow \epsilon \text{ est}(Q) \stackrel{1}{\exists}$	$\epsilon \text{ est}(Q) \sqsubseteq Q^\circ \text{ — (7.1.1a)}$
(b)	$\sqsubseteq Q^\circ \stackrel{1}{\exists}$	$\stackrel{1}{\exists} \sqsubseteq \epsilon \text{ — } T^\circ \stackrel{T}{\exists} \sqsubseteq \epsilon$
(c)	$\sqsubseteq Q^\circ \epsilon$	

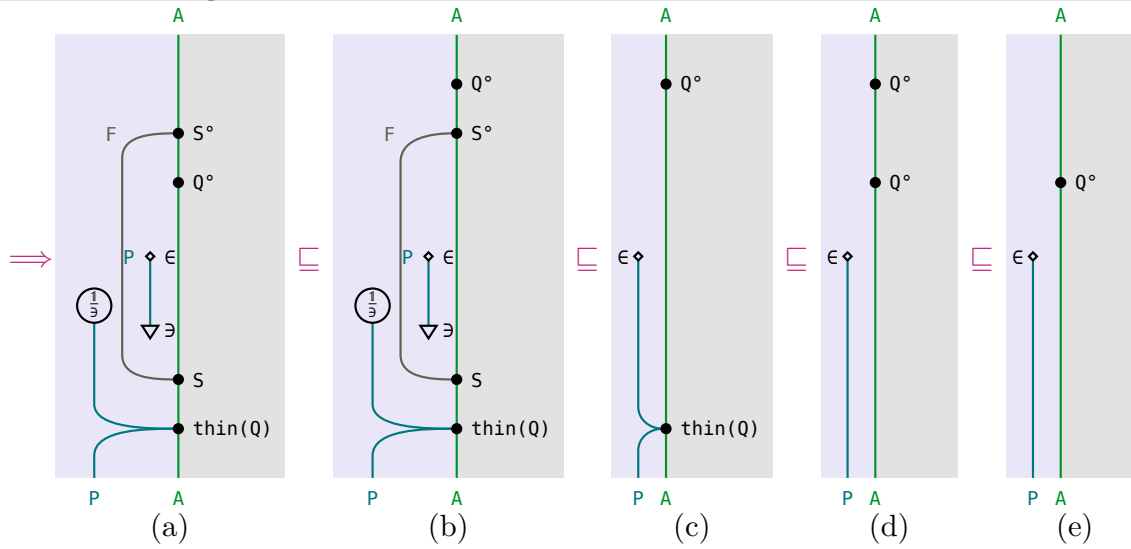
$$Q \subseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

$$\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \exists \langle (S) \rangle \quad \text{and} \quad \langle (S) \rangle \circ \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \subseteq Q^\circ \in$$

(8.1a.1) at $X \triangleq \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle$, $\langle (S) \rangle$ for its S

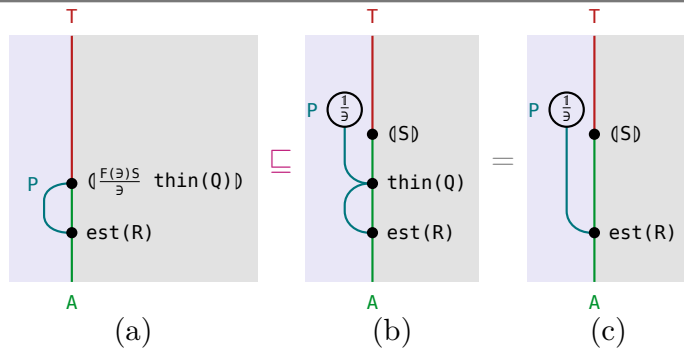


(a)	$\Rightarrow S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$S^\circ F(Q^\circ) \subseteq Q^\circ S^\circ \text{ — (7.2b), (5.1b.3)}$
(b)	$\sqsubseteq Q^\circ S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \subseteq \epsilon \text{ — } T^\circ \frac{T}{\exists} \subseteq \epsilon$
(c)	$\sqsubseteq Q^\circ \in \text{thin}(Q)$	$\epsilon \text{thin}(Q) \subseteq Q^\circ \in \text{ — } \triangle \text{thin, (0.9a.4)}$
(d)	$\sqsubseteq Q^\circ Q^\circ \in$	$Q \text{ a preorder}$
(e)	$\sqsubseteq Q^\circ \in$	

$$Q \subseteq R \wedge 1 \subseteq Q \wedge Q \subseteq Q \wedge R \subseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$$

Corollary 8.1

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \subseteq R$, both preorders



(a)	$\langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \text{est}(R)$	Theorem 8.1 $Q \subseteq Q \wedge \text{MonoAlg}(S, Q)$ $\Rightarrow \langle \frac{F(\exists) S}{\exists} \text{thin}(Q) \rangle \subseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$
(b)	$\sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle \text{est}(R)$	$Q \subseteq R \wedge 1 \subseteq Q \wedge R \subseteq R$ $\Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R)$
(c)	$= \langle \frac{(S)}{\exists} \text{est}(R) \rangle$	

§8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$.

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$Q \triangleq R \cap (\text{head} \text{head}^\circ)$,

$\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$,

$\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$.

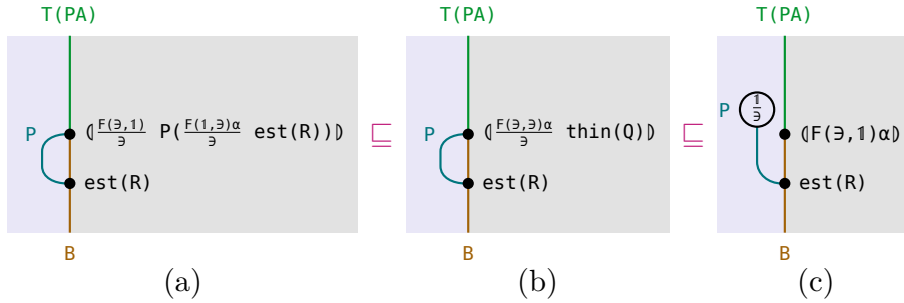
(8.2b)

the law	what it says
1 $F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
2 $S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	S head is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge R \cap ((F(1, \exists) \alpha)^\circ F(1, \exists) \alpha) \sqsubseteq Q \quad (8.2c)$$

$$\Rightarrow \left(\frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \right) \text{est}(R) \sqsubseteq \left(\frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$$

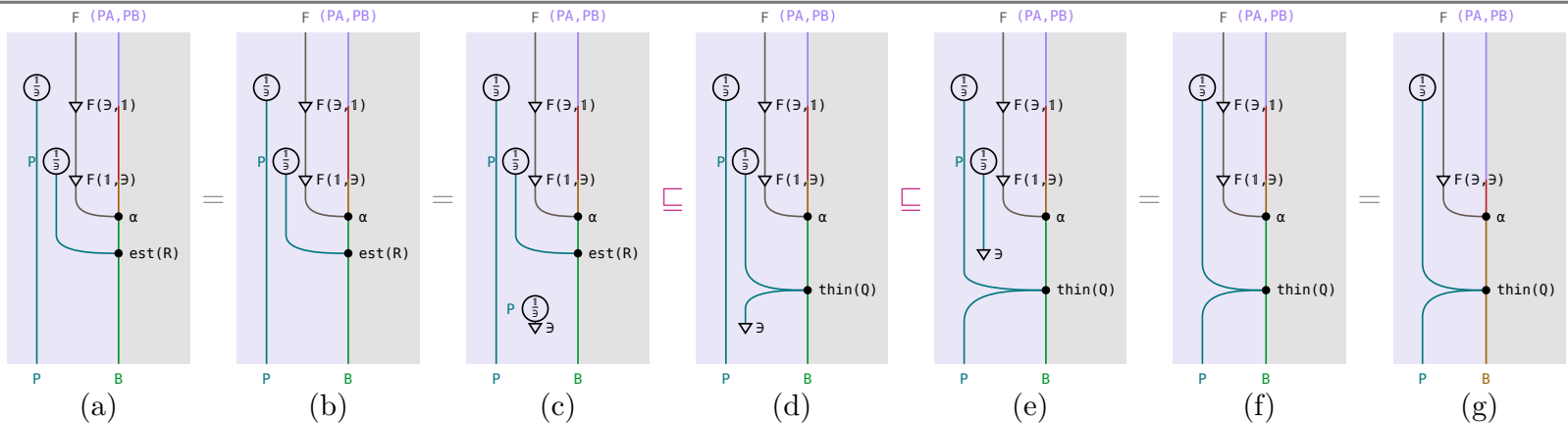
a least-cost path in a layered network, as a fold over the layers



(a)	$\left(\frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \right) \text{est}(R)$	(8.2d) $R \cap (S^\circ S) \sqsubseteq Q$ $\Rightarrow \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$
(b)	$\sqsubseteq \left(\frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \right) \text{est}(R)$	$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge$ $\text{MonoAlg}(F(\exists, 1) \alpha, Q)$
(c)	$\sqsubseteq \left(\frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$	$\Rightarrow \left(\frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \right) \text{est}(R) \sqsubseteq \left(\frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$

$$S \triangleq F(1, \exists) \alpha, R \cap (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \sqsubseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



(a)	$\frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right)$	
(b)	$= \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right)$	$P \left(\frac{1}{\exists} \right) E(\exists) = 1$ $P(R \ S) = P(R) P(S)$
(c)	$= \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \frac{1}{\exists} E(\exists)$	(8.3) $R \cap (S^\circ S) \sqsubseteq Q$ $\Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{thin}(Q)$ $P(R \ S) = P(R) P(S)$
(d)	$\sqsubseteq \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \right) P(\text{thin}(Q)) E(\exists)$	$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q)$
(e)	$\sqsubseteq \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \right) E(\exists) \text{thin}(Q)$	$\frac{V \ S}{\exists} = \frac{V}{\exists} P \left(\frac{S}{\exists} \right) \text{ union}$
(f)	$= \frac{F(\exists, 1) F(1, \exists) \alpha}{\exists} \text{thin}(Q)$	$F(R, 1) F(1, S) = F(R, S)$
(g)	$= \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$	

$$= \text{cp}(F) P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}]$$

at the layered network $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}])$: $\frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \triangleq \text{cost}$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangleq \text{cost}$$

$$\begin{array}{ccccc}
 F(A, X) & = & A & + & A \times X \\
 & & | & & | \quad | \\
 & & R & & R \quad S \\
 & & \downarrow & & \downarrow \quad \downarrow \\
 F(A', X') & = & A' & + & A' \times X'
 \end{array}$$

(8.2e)

u $F(R, S)$ v holds iff one of:

- u, v are both inl and d R d': the leaf moves by R;

- u, v are both inr and $p_1 \ R \ q_1 \wedge p_2 \ S \ q_2$: the vertex moves by R , the rest by S .
- An inl never relates to an inr : $R+S \triangleq [R \ \text{inl}, S \ \text{inr}]$ returns to the summand it came from, and $F(1)=1$ allows no crossing.
- $F(\exists, \exists)$ relates $\text{inr}(\{a, b\}, \{p, q\})$ to $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$, and $\text{inl}(\{a, b\})$ to $\text{inl}(a), \text{inl}(b)$.

§8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow PA$, $\text{cup} : PA \times PA \rightarrow PA$, $\text{cp}(F) \text{ cp}(F) \triangleq \frac{F(\exists)}{3}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$ for $\leq$ a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $Q[\text{nil}, \text{bump } Q]$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

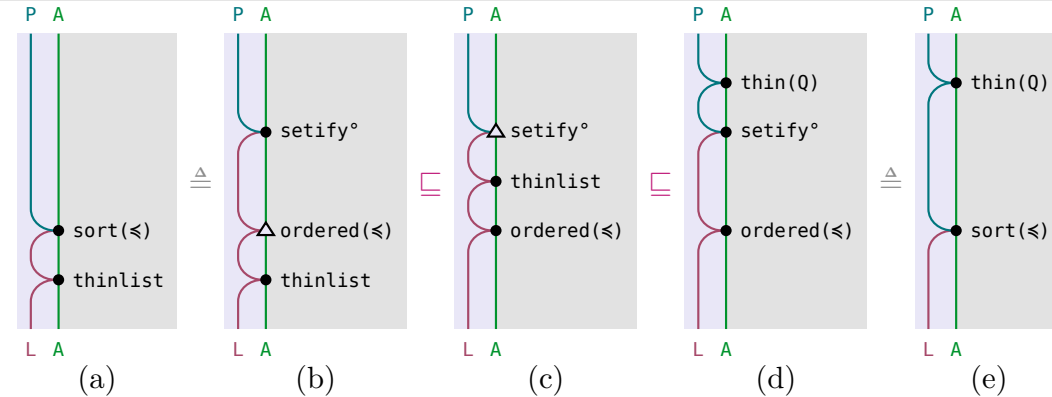
Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; $\leq$ a connected preorder with both f_1, f_2 monotonic on $\leq$; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

(8.3b)

the law	what it says
1 $\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
2 $\text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6) — (8.6) $\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
3 $\text{sort}(\leq) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
4 $\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$ (8.8)	shunt a function through a sort
5 $\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
6 $(\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)	merging two sorted lists sorts their union
7 $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

$\text{Map}(\text{setify}) \wedge \text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist setify} \sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$
a thinning of the sorted list lists a thinning of the set — $\leq$ a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.

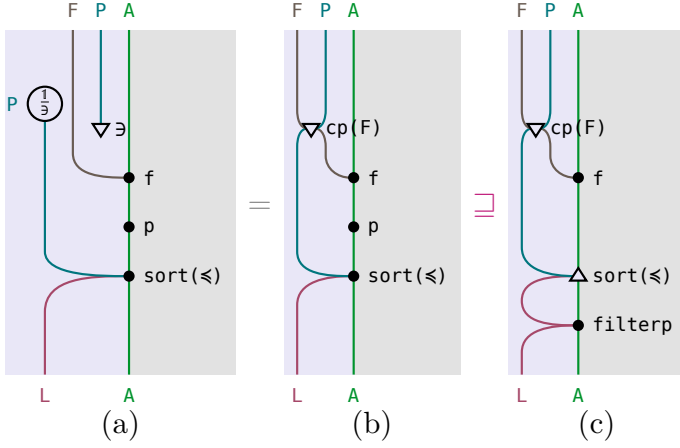
(8.6)



(a)	$\text{sort}(\leq) \text{ thinlist}$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$
(b)	$\triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ thinlist}$	$\text{Coreflexive}(\text{ordered}(\leq)) \wedge g \sqsubseteq \text{subseq} \wedge \text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \Rightarrow \text{setify}^\circ \text{ ordered}(\leq) g \sqsubseteq \text{setify}^\circ g \text{ ordered}(\leq)$
(c)	$\sqsubseteq \text{setify}^\circ \text{ thinlist ordered}(\leq)$	$\text{Map}(\text{setify}) \wedge g \text{ setify} \sqsubseteq \text{setify } T \Rightarrow \text{setify}^\circ g \text{ ordered}(\leq) \sqsubseteq T \text{ setify}^\circ \text{ ordered}(\leq)$
(d)	$\sqsubseteq \text{thin}(Q) \text{ setify}^\circ \text{ ordered}(\leq)$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$
(e)	$\triangleq \text{thin}(Q) \text{ sort}(\leq)$	

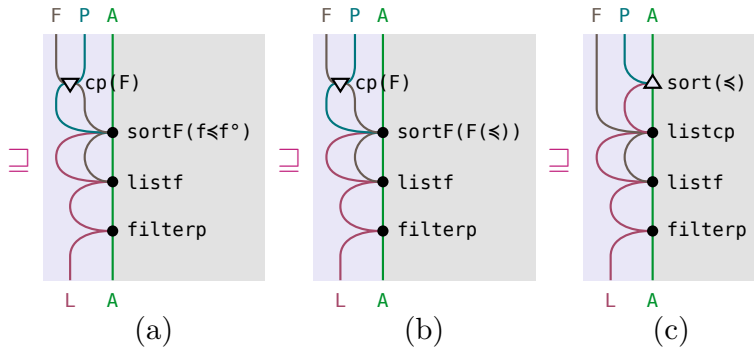
Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$) $\wedge$ MonoAlg(f, $\leq$) $\wedge$ sortF(f $\leq$ f $^\circ$) listf $\sqsubseteq$ Lemma 8.1
P(f) sort($\leq$) $\wedge$ sort($\leq$) filterp $\sqsubseteq$ E(p) sort($\leq$) $\wedge$ F(sort($\leq$)) listcp $\sqsubseteq$ cp(F) sortF(F($\leq$))
 $\Rightarrow$ F(sort($\leq$)) listcp listf filterp $\sqsubseteq$ $\frac{F(\exists) f p}{\exists}$ sort($\leq$)
 one sorted list built from sorted arguments, instead of a set built and then sorted — f : FA $\rightarrow$ A monotonic
 on $\leq$, p coreflexive, F linear.

Map(f) $\wedge$ sort($\leq$) filterp $\sqsubseteq$ E(p) sort($\leq$) $\Rightarrow$ cp(F) P(f) sort($\leq$) filterp $\sqsubseteq$ $\frac{F(\exists) f p}{\exists}$ sort($\leq$)
 sorting the f-images of the product and filtering by p is below sorting the set F($\exists$)fp builds



(a)	$\frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$	$\frac{F(\exists) f p}{\exists} = \frac{F(\exists)}{\exists} E(fp) \sqsubseteq \frac{S}{\exists} E(R) = \frac{S R}{\exists}, \triangleq \text{thinlist}$ $E(f) = P(f), f \text{ a map} \rightarrow \text{Map}(f)$ $\Rightarrow P(f) = E(f)$
(b)	$= \text{cp}(F) P(f) E(p) \text{ sort}(\leq)$	$\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq) \rightarrow (8.3b.5)$
(c)	$\sqsubseteq \text{cp}(F) P(f) \text{ sort}(\leq) \text{ filterp}$	$\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq) \rightarrow (8.3b.4)$

Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$) $\wedge$ MonoAlg(f, $\leq$) $\wedge$ sortF(f $\leq$ f $^\circ$) listf $\sqsubseteq$ P(f) sort($\leq$) $\wedge$
F(sort($\leq$)) listcp $\sqsubseteq$ cp(F) sortF(F($\leq$)) $\Rightarrow$ F(sort($\leq$)) listcp listf filterp $\sqsubseteq$ cp(F) P(f) sort($\leq$) filterp
 combining the arguments' sorted lists by listcp, mapping f and filtering is below sorting the f-images and filtering



(a)	$\sqsubseteq \text{cp}(F) \text{ sortF}(f \leq f^\circ) \text{ listf filterp}$	$F(\leq) \sqsubseteq f \leq f^\circ \rightarrow (7.2b), \triangleq \text{thinlist}$
(b)	$\sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \text{ listf filterp}$	$F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq)) \rightarrow (8.3b.7)$
(c)	$\sqsubseteq F(\text{sort}(\leq)) \text{ listcp listf filterp}$	

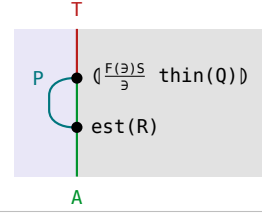
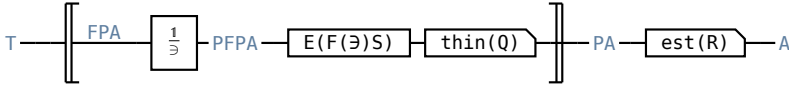
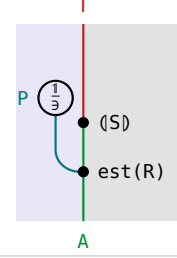
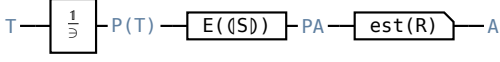
Theorem 8.2

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1, p_1, Q) \wedge$
 $\text{MonoAlg}(f_2, p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge$
 $\text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq$
 $P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq)$
 $\wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq)$
 $\wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = (f_1 p_1) \cup (f_2 p_2) \Rightarrow (\text{listcp} \langle g_1, g_2 \rangle \text{ merge}(\leq))$
 $\text{thinlist}(Q) \sqsubseteq \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \text{ est}(R)$

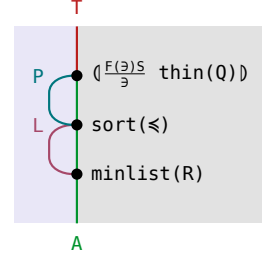
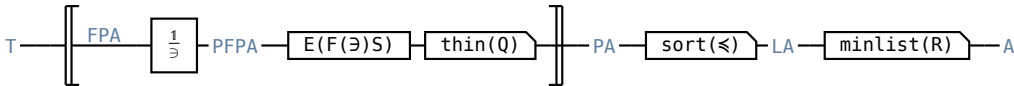
a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

circuit

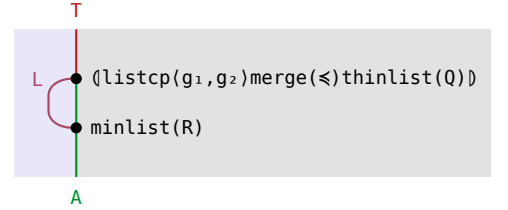
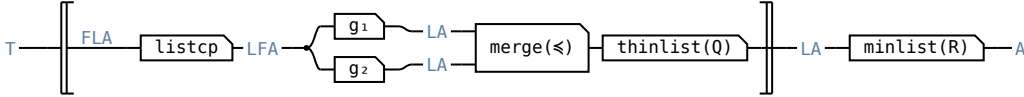
Hinze–Marsden



$\sqsubseteq$ Corollary 8.1 $Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \sqsubseteq R \wedge \text{MonoAlg}(S, Q)$
 $\Rightarrow (\frac{F(3)S}{3} \text{ thin}(Q)) \text{ est}(R) \sqsubseteq \frac{(S)}{3} \text{ est}(R)$ at $f_1 p_1$ and $f_2 p_2$ monotonic on Q —
 $\triangleq \text{thinlist}$



$\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R)$ — (8.3b.3) at its $Q \triangleq R$



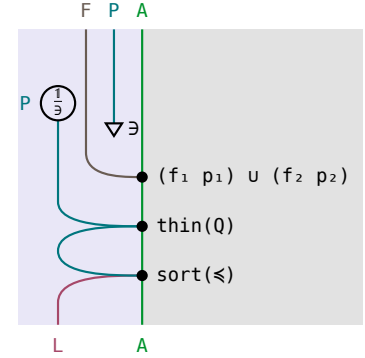
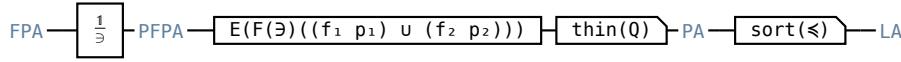
fusion at (8.3c) $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1$
 $\sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq)$
 $\text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge$
 $F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq))$
 $\text{merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$
 $\Rightarrow F(\text{sort}(\leq)) \text{ listcp} \langle \text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2 \rangle \text{ merge}(\leq)$
 $\text{thinlist}(Q) \sqsubseteq \frac{F(3) ((f_1 p_1) \cup (f_2 p_2))}{3} \text{ thin}(Q) \text{ sort}(\leq)$

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{ listcp} (\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)
\end{aligned}$$

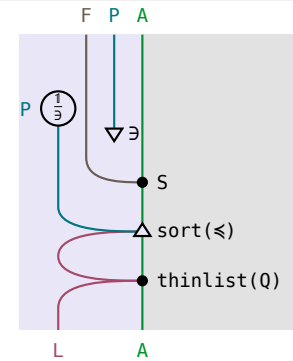
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{FPA} \rightarrow \left[\frac{1}{3} \right] \rightarrow \text{PFPA} \rightarrow \text{E}(F(\exists)S) \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \quad \text{--- (8.3b.2)}
\end{aligned}$$



$$\begin{aligned}
& \text{FPA} \rightarrow \left[\frac{1}{3} \right] \rightarrow \text{PFPA} \rightarrow \text{E}(F(\exists)f_1 p_1) \rightarrow \text{PA} \rightarrow \text{cup} \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \\
& \left(\frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \right) \text{ cup} \text{ sort}(\leq) \text{ thinlist}(Q) \\
& S = (f_1 p_1) \cup (f_2 p_2) \quad \text{---} \triangleq \text{thinlist, then} \triangleq \text{cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow \left[\frac{1}{3} \right] \rightarrow \text{PFPA} \rightarrow \text{E}(F(\exists)f_1 p_1) \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \\
& \left(\frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \right) (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q) \\
& (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \quad \text{--- (8.3b.6)}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow \left[\frac{1}{3} \right] \rightarrow \text{PFPA} \rightarrow \text{E}(F(\exists)f_1 p_1) \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{LA} \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \\
& \left(\frac{F(\exists)f_1 p_1}{\exists} \text{ sort}(\leq), \frac{F(\exists)f_2 p_2}{\exists} \text{ sort}(\leq) \right) \text{ merge}(\leq) \text{ thinlist}(Q) \\
& \langle X, Y \rangle (\text{sort}(\leq) \times \text{sort}(\leq)) = \langle X \text{ sort}(\leq), Y \text{ sort}(\leq) \rangle \quad \text{--- absorption}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow F(\text{sort}(\leq)) \rightarrow \text{FLA} \rightarrow \text{listcp} \rightarrow \text{LFA} \rightarrow \left(\begin{array}{c} \text{listf}_1 \quad \text{filterp}_1 \\ \text{listf}_2 \quad \text{filterp}_2 \end{array} \right) \rightarrow \text{LA} \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& F(\text{sort}(\leq)) \text{ listcp} \langle g_1, g_2 \rangle \text{ merge}(\leq) \text{ thinlist}(Q) \\
& \text{Lemma 8.1 Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, \leq) \wedge \\
& \text{sortF}(f \leq f^\circ) \text{ listf} \sqsubseteq P(f) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp} \sqsubseteq E(p) \text{ sort}(\leq) \wedge \\
& F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \\
& \Rightarrow F(\text{sort}(\leq)) \text{ listcp} \text{ listf} \text{ filterp} \sqsubseteq \frac{F(\exists) f p}{\exists} \text{ sort}(\leq) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then} \\
& X \langle g_1, g_2 \rangle \sqsubseteq \langle X g_1, X g_2 \rangle \quad \text{---} X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle ; g_i \triangleq \text{list}(f_i) \text{ filter}(p_i) \quad \text{---} \triangleq \\
& \text{thinlist}
\end{aligned}$$

§8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$, $\text{value} \triangleq \text{list}(\text{vol}) \text{ sum}$, $\text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}$.

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2])$, within w the coreflexive on xs with weight $\text{xs} \leq w$, $0 \leq w$.

$R \triangleq \text{value} \geq \text{value}^\circ$, $Q \triangleq \text{weight} \leq \text{weight}^\circ$, $P \triangleq R$,.

$\text{FA} = 1 + \text{Item} \times A$, $\text{listcp} = \text{wrap} + \text{cpr}$, $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])$ $\text{filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$,
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$.

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$, $h_2 \triangleq \text{list}(\pi_2)$.

(8.4b)

the law	what it says
1 $(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R : a selection of greater value need not still fit once one more item goes in
2 $(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R$ (8.4c)
 $[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sort}(R) \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq$
 $P([\text{nil}, \pi_2]) \text{sort}(R) \wedge \text{sort}(R) \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sort}(R) \wedge F(\text{sort}(R)) \text{listcp} \sqsubseteq \text{cp}(F)$
 $\text{sortF}(F(R)) \wedge (\text{sort}(R) \times \text{sort}(R)) \text{merge}(R) \sqsubseteq \text{cup } \text{sort}(R) \wedge \text{sort}(R) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt}))$
 $\text{sort}(R) \wedge \text{sort}(R) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp } (\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \text{ merge}(R) \text{ thinlist})$
 $\text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supset} \text{est}(R)$

the knapsack problem, as a fold that thins the packings kept at each item

circuit	Hinze–Marsden
$\text{subseq } (\text{within } w) \supset \text{est}(R)$	
$\frac{[(\text{nil}, \text{cons}) (\text{within } w) \cup [\text{nil}, \pi_2]]}{\supset} \text{est}(R)$ fusion, weights non-negative.	
$(\text{listcp } (g_1, g_2) \text{ merge } R \text{ thinlist}(Q)) \text{ minlist}(R)$ binary thinning theorem, at $P \triangle R$, F linear, Q from (8.4b.2)	
$[\text{nil}, \text{cpr } (h_1, h_2) \text{ merge } R \text{ thinlist}(Q)] \text{ minlist}(R)$ $= \text{listcp} = \text{wrap} + \text{cpr}$, $g_i = [\text{list}(\text{nil}), h_i] \rightarrow \triangleq \text{within}$; $\text{minlist}(R)$ is head, packings coming out in descending value	

§8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.
width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.
white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.
R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .
g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

	the law	what it says
1	(1×R) glue⊆glue R FALSE	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
2	(1×Q) new⊆new Q (1×Q) (glue (ok w))⊆glue (ok w)Q cons monotonic on collect≤collect°.	both halves are monotonic on Q once ties in waste are broken by the first line
3	merge T=cat; P≐head prefix head° also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

($\forall a. 0 \leq \text{len}(a) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap}$ (8.5c)
 $\text{wrap}, \text{new}] \sqsubseteq [\text{wrap wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap wrap}, \text{new}]) \text{sort}(\tau) \wedge \text{sortF}([\text{wrap wrap}, \text{glue}] \sqsubseteq \tau$
 $[\text{wrap wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap wrap}, \text{glue}]) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{filterp}_2 \sqsubseteq E(\text{ok}(w)) \text{sort}(\tau)$
 $\wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{merge}(\tau) \sqsubseteq \text{cup} \text{sort}(\tau) \wedge$
 $\text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}$
 $(\text{listf}_1, \text{listf}_2 \text{filterp}_2) \text{merge}(\tau) \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\supset} \text{est}(R)$
 a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E(\text{partition fits}(w))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\frac{\text{partition list}^*(\text{fits } w)}{\supset} \text{est}(R)$	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{Q([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))}{\supset} \text{est}(R)$ fusion, every word fits on a line by itself.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{Q([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))}{\supset} \text{est}(R)$ the algebra as $(f_1 p_1) \sqcup (f_2 p_2)$, $p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist}$.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\sqsubseteq ([\text{listcp} (g_1, g_2) \text{cat thinlist}(Q)] \text{minlist}(R))$ binary thinning theorem, at $P \triangleq T$ with merge $\tau = \text{cat}$ ((8.5b.3)), Q from (8.5b.2)	
$Q([\text{start}, \text{cpr} (h_1, h_2) \text{cat thinlist}(Q)] \text{minlist}(R))$ $= \text{listcp} = \text{wrap} + \text{cpr}, g_i \text{ along the coproduct — } \triangleq \text{partition}$	

§8.6. Bitonic tours

(8.6a)

$FA = (\text{City} \times \text{City}) + (\text{City} \times A)$, the base functor of cons-lists of length at least two; $\text{listcp} = \text{wrap} + \text{cpr}$; $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$, neither positive nor symmetric.

$\text{start } (a, b) = ([a, b], [a, b])$, $\text{dropl } (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{dropr } (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $\text{tour} \triangleq ([\text{start}, \text{dropl} \sqcup \text{dropr}])$.

$\text{cost } (xs, ys) = \text{outcost } xs + \text{incost } ys$, $\text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n)$, $\text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1})$.

$\text{next} \triangleq \text{tail head}$, $\text{next2} \triangleq \text{next} \times \text{next}$, $\text{head2} \triangleq \text{head} \times \text{head}$, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $Q \triangleq R \circ (\text{next2 next2}^\circ) \circ (\text{head2 head2}^\circ)$, $P \triangleq T$, $g_1 \triangleq \text{list}([\text{start}, \text{dropl}])$, $g_2 \triangleq \text{list}([\text{start}, \text{dropr}])$.

(8.6b)

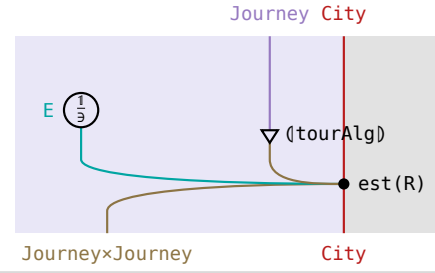
the law	what it says
1 $(1 \times R) \text{dropl} \sqsubseteq \text{dropl } R$ FALSE $(1 \times R) \text{dropr} \sqsubseteq \text{dropr } R$ FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
2 $(1 \times Q) \text{dropl} \sqsubseteq \text{dropl } Q$ $(1 \times Q) \text{dropr} \sqsubseteq \text{dropr } Q$,	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

$$\begin{aligned}
& (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{start}, \text{droplFn}] \tau [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq \quad (8.6c) \\
& P([\text{start}, \text{droplFn}]) \text{sort}(\tau) \wedge \text{sortF}([\text{start}, \text{droprFn}] \tau [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq \\
& P([\text{start}, \text{droprFn}]) \text{sort}(\tau) \wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{cat} \\
& \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Qc(tc)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \implies (\text{listcp} \\
& \quad (g_1, g_2) \text{cat thinlist}) \text{minlist} \sqsubseteq \frac{(\text{tourAlg})}{\exists} \text{est}(R) \\
& \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city}
\end{aligned}$$

circuit

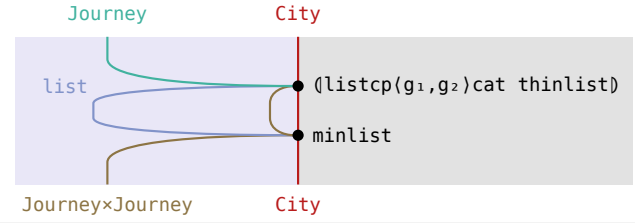
Hinze–Marsden

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} P(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\
& \frac{\text{tour}}{\exists} \text{est}(R)
\end{aligned}$$



$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \cup [\text{start}, \text{droprFn}])} P(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\
& = \frac{Q([\text{start}, \text{dropl} \cup \text{dropr}])}{\exists} \text{est}(R) \\
& \text{tour} \triangleq Q([\text{start}, \text{dropl} \cup \text{dropr}]) \triangleq \text{tour}
\end{aligned}$$

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{\text{listcp}} \text{list}(F(\text{Journey}(\text{City})^2)) \xrightarrow{\text{cat}} \text{thinlist} \xrightarrow{\text{minlist}} \text{Journey}(\text{City}) \\
& \sqsubseteq Q(\text{listcp}(g_1, g_2) \text{cat thinlist}(Q)) \text{minlist}(R) \\
& \text{binary thinning theorem, at } P \triangleq T \text{ with merge } \tau = \text{cat}, Q \text{ from (8.6b.2)}
\end{aligned}$$



$$\begin{aligned}
& Q([\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\
& = \text{thinlist}(Q)] \text{minlist}(R) \\
& \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\
& \text{quadratic, two tours added per step}
\end{aligned}$$